

Lecture 3

Cardinality and Related Terms

- For a set S , $\text{card}(S) = \#(S) = |S| = \text{number of element in } S$
- Finite: $S = \emptyset$ or $\exists n \in \mathbb{N}$ & bijection $h : \{1, 2, \dots, n\} \rightarrow S$
- Denumerable: $\exists$ bijection $h : \mathbb{N} \rightarrow S$
- Countable: finite or denumerable
- Uncountable: none of the above
- Bijection implies same cardinality
- Denumerability means $\exists$ a list of elements of S : $x_1, x_2, x_3, \dots$

$\mathbb{Q}$ is denumerable

- Start with $\mathbb{Q}^+$
- Write out the positive rationals in a grid, each row has the same denominator
- A path along diagonal lines covers all terms
- Thus, $\mathbb{Q}^+ \sim \mathbb{N}$
- Similarly, $\mathbb{Q}^- \sim \mathbb{N}$
- Alternate values from $\mathbb{Q}^+$ and $\mathbb{Q}^-$ to get $\mathbb{Q}$, $\mathbb{Q} \sim \mathbb{N}$

$\mathbb{Q} \times \mathbb{Q}$

- Cartesian product of two denumerable sets is denumerable
- Pairs of elements of denumerable sets
- If infinite $\mathbb{Q} \times \mathbb{Q} \times \dots$, then it's not necessarily denumerable

$A \cup B$

- 2 infinite set A and B
- Same cardinality as larger set

$[0, 1]$ is uncountable

- Every real value has a decimal expansion
- Suppose the interval is countable
- List out all the values in the interval $x_1, x_2, x_3, \dots$
 - Decimal representation of x_i is $0.x_{i1}x_{i2}x_{i3}\dots$
- Now we must create a number that isn't included in this list
- Let $y_i \neq x_{ii}$ and $y_i \neq 9$
 - Ensures that we don't double count values like 0.01 and 0.0999...
- $\forall x_i, x_{ii} \neq y_i$, so y isn't in the list
- Exercise: can try using binary expansions

$\mathbb{Q} \times [0, 1]$

- Uncountable as any fixed rational already makes it uncountable

$[0, 1]$ to $[a, b]$

- Bijection exists, $T : t \rightarrow a + (b - a)t$
- How does this relate to the set of points that makes up a circle?
 - Think about the interval being wrapped around

$[0, 1]$ vs. $(0, 1]$

- Can use the idea of Hilbert hotel
- Hilbert hotel deals with a countable number of rooms, so we look at a countable subset of $(0, 1]$

- We shift all these values down 1 room
 - Now created space to include 0 into the bijection
- Values not in the countable subset are mapped to themselves

Schroeder-Bernstein Theorem

Historical background

- Schroeder gave a wrong proof of the theorem
- Bernstein gave first published proof
 - Dedekind had an unpublished proof
- Also called Cantor-Bernstein Theorem
 - May be more apt name, as Cantor had already stated it without proof

Theorem Statement

- For two sets A and B , if there exists 2 injections f, g from $A \rightarrow B$ and $B \rightarrow A$, respectively, there exists a bijection between A and B .
- Similar idea to showing that when $a \leq b$ and $b \leq a$, $a = b$

Proof 1: Diagram

- Unfortunately I can't draw diagrams
- Two circles/regions representing A and B
 - A subregion is a concentric circle inside
- Subregion inside A representing gB , and subregion inside B representing fA
- $(gf)A$ is a subregion of gB and $(fg)B$ is a subregion of fA
- $gB \setminus (gf)A \sim B \setminus fA$ and $fA \setminus (fg)B \sim A \setminus gB$
- Already injection, must show surjection
- For a value to be in gB , there must be some b such that $g(b)$ is defined
 - Similar for fA
- Thus, $A \setminus (gf)A$ and $B \setminus (fg)B$ are bijections
- Continue this process into $(gf)A$ and $(fg)B$, sum all the rings
- Book stops here, Pugh adds some additional comments in class
 - Something about points in $A \setminus gB$
 - Not sure what he means though, I don't see an issue with this proof

Proof 2: Orbits

- Suppose $a \in A$
- The set of orbits consists of elements from $\{(gf)^n(a) : n \geq 0 \text{ or } n \text{ is any negative integer}\}$
- Start with an element a , call it a_0
- We can start applying gf to A
 - $(gf)(a) = a_1, (gf)^2(a) = a_2, \dots, (gf)^n(a) = a_n$
- 3 types of orbits
 - Think about going backwards from a
 - Unidirectional: has a starting point a^{-k}
 - Bijection can be shown by mapping starting points
 - Bidirectional: goes back infinitely
 - Think about map of mapping the natural numbers to the value that is 1 greater
 - Periodic
 - There's a cycle in the values
 - Can have period 1, 2, 3, etc.
 - Can map orbits of the same period

Cantor Theorem

- Power set of A is the collection of all subsets, known as $\mathcal{P}(A)$
- $\forall A (|A| < |\mathcal{P}(A)|)$
- Ex. $\emptyset$ has no elements, but the power set of it has 1

Consequence

- $|\mathcal{P}(A)| < |\mathcal{P}(\mathcal{P}(A))|, |\mathcal{P}(\mathcal{P}(A))| < |\mathcal{P}(\mathcal{P}(\mathcal{P}(A)))|$, etc.
- Inequality stays true when nesting
- Shows there is no longest infinity as you can keep applying the power set
- Groundbreaking discovery